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# MORGAN PFAFF

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❖ [morgan.pfaff1@gmail.com](mailto:morgan.pfaff1@gmail.com) ❖ [linkedin.com/in/morgan-pfaff](https://www.linkedin.com/in/morgan-pfaff) ❖ [morganpfaff.github.io](https://morganpfaff.github.io) ❖

## EDUCATION

### University of California (UC) Berkeley and UC San Francisco

Berkeley, CA

Doctor of Philosophy in Bioengineering

October 2026

- Dissertation: “Viscoelastic, Semi-synthetic HyA Hydrogels for Soft Tissue Regeneration”
- Advisor: Kevin E. Healy, PhD
- National Science Foundation Graduate Research Fellow
- GPA 4.0/4.0
- Major area: Biomaterials Engineering & Tissue Regeneration; Minor area: Cell, Tissue, & Developmental Biology

### Northeastern University

Boston, MA

Bachelor of Science in Biochemistry

May 2021

- GPA 4.0/4.0
- Honors Program; Honors in the Discipline
- Engineering & entrepreneurship study abroad program, Oxford, United Kingdom (Summer 2019)

## RESEARCH EXPERIENCE

### Kevin Healy Lab, UC Berkeley

Berkeley, CA

PhD Candidate

January 2022 – August 2026

- Engineered and mechanically characterized tunable, viscoelastic, semi-synthetic hyaluronic acid (HyA) hydrogels to promote muscle regeneration after rotator cuff injury and volumetric muscle loss; in a murine delayed rotator cuff repair model, the hydrogel improved recovery of muscle quality and vascularization
- Investigated hydrogel-driven mechanisms of muscle regeneration, including directed differentiation of fibro-adipogenic progenitor cells (FAPs) and associated fusion of muscle stem cells, in 3D *in vitro* cultures and cocultures
- Performed single-cell RNA-seq analysis of patient-derived FAPs and used the findings to guide hydrogel design, including degradation rate, ligand presentation, drug encapsulation and release kinetics, and injectability
- Prepared hydrogels for, and analyzed and interpreted data from, *in vivo* injury studies conducted with orthopedic surgery collaborators at UCSF and UVA
- Published 2 peer-reviewed papers, gave 5 presentations at 4 national and international conferences, and co-invented part of the hydrogel technology now in development at MuscleMatrix
- Mentored 3 PhD rotation students, 1 visiting international PhD student, 1 undergraduate, and a team of 4 Master of Translational Medicine students, and trained dozens of other students on equipment and methods

### Randall Erb Lab, Northeastern University

Boston, MA

Biomaterials Undergraduate Researcher

September 2016 – July 2021

- Developed a fabrication protocol for dual-network hydrogels reinforced with magnetically responsive hydroxyapatite fibers for bone graft applications, and characterized their mechanics using dynamic mechanical analysis and scanning electron microscopy
- Designed and analyzed COMSOL finite element simulations of hydroxyapatite fiber rotation in an alginate matrix to investigate potential new toughening mechanisms associated with composite movement
- Fabricated freeze-cast collagen conduits with aligned, double-layered porosity for peripheral nerve repair in collaboration with Dartmouth College and Harvard Medical School
- Secured independent research funding by writing a successful proposal for the competitive Project-Based Exploration for the Advancement of Knowledge (PEAK) Summit Award

## **Marc Hansen Lab, University of Connecticut Health Center**

**Farmington, CT**

Summer Research Fellow

Summers 2017 & 2018

- Piloted synthetic-lethal experiments testing inhibitor and siRNA combinations in osteosarcoma cell lines and hFOB cells to identify druggable vulnerabilities in growth factor receptor signaling, using PCR and absorbance assays
- Used laser capture microdissection on histological sections to isolate osteoblast and osteoclast populations and investigate cell behavior and genetics in Paget's disease of bone
- Presented results to physicians and scientists at UConn Health; work provided preliminary data for the lab's NIH grant applications

## **TEACHING EXPERIENCE**

### **University of California, Berkeley**

**Berkeley, CA**

Reader

Fall 2026

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy; 33 students): grade homework, exams, and team design projects, including multiple written reports and presentations
- BIOE 252: Clinical Need-Based Therapy Solutions (Instructor: Dr. Syed Hossainy; ~100 students): form and guide student teams developing therapy concepts for unmet clinical needs, grade homework and team projects, and coordinate guest lecturers from around the country

Guest Lecturer

2023 – 2026

- Delivered 4 full-length lectures in BIOE/MSE C118/C208 (two on hydrogels, one on polymer viscoelasticity, and an introduction to polymers) in place of the course instructor, Dr. Kevin Healy

Graduate Student Instructor (GSI)

Fall 2022

- BIOE/MSE C118/C208: Biological Performance of Materials (Instructor: Dr. Kevin Healy)
- Led a weekly discussion section reviewing lecture material and held weekly office hours for a class of ~40 students
- Administered and graded homework and exams, and evaluated ~10 team design projects, including 15-page reports proposing biomaterial solutions to unmet clinical needs and a series of pitch, 10-minute, and 20-minute presentations

## **TRANSLATIONAL, REGULATORY, & INDUSTRY EXPERIENCE**

### **MuscleMatrix**

**Berkeley, CA**

Scientific Consultant

August 2026 – Present

- Consult for an early-stage startup spun out in part from my doctoral research that is developing an injectable, cell-free hyaluronic acid scaffold to support muscle regeneration after rotator cuff injury
- Help build company operations from the ground up, including regulatory compliance, environmental health and safety (EHS), laboratory buildout and equipment procurement, and manufacturer and supplier sourcing
- Contribute to FDA regulatory strategy for approval as a medical device or combination product, and write grant applications to secure funding
- Conduct formulation development, characterizing hydrogel injectability and mechanical properties and evaluating how drug loading alters material properties to inform a planned drug-loaded version
- Train new scientific staff on the hydrogel platform, complex technical procedures, and laboratory operations

### **Vertex Pharmaceuticals**

**Boston, MA**

Exploratory Research Co-op

January 2020 – June 2020

- Developed and optimized a reporter assay to detect transcription factors in potentially druggable pathways for sickle cell disease (SCD), giving the team a faster, more accurate readout of compound effectiveness
- Generated stable cell lines by molecular cloning of recombinant vectors for genomic integration
- Independently planned, executed, and analyzed experiments, and presented results and journal club articles to the SCD exploratory research team

- Screened sgRNAs to identify the most effective candidates for the company's pipeline diseases, introducing CRISPR-mediated edits via nucleofection, transfection, and viral transduction
- Programmed and optimized high-throughput screening assays on automated liquid handlers, including a comparative performance evaluation of the Biomek i7 and its Biomek FX predecessor, enabling more complex, higher-throughput sgRNA screening
- Analyzed data in Prism, Tableau, and FlowJo and presented findings to the screening group and other departments

## **PUBLICATIONS & PRESENTATIONS**

### **Publications**

1. Wang, V. Y., Zhu, Y., Spaulding, K. A., **Pfaff, M. R.**, Neiman, G., Takaba, K., Chen, N., Parekh, K., Hasaballa, A., Haraldsson, H., Baker, A., Saloner, D., Wallace, A. W., Ziats, N. P., Healy, K. E., Ratcliffe, M. B. NIPAAm-co-PEG4000 thermoresponsive hydrogel with ROS scavenging properties improves in vivo border zone contractility and reduces myocardial remodeling in sheep after MI. *Acta Biomaterialia*, 2026, 211, 48–60. doi.org/10.1016/j.actbio.2025.06.046
2. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. *Advanced Healthcare Materials*, 2025, 14, 2403962. doi.org/10.1002/adhm.202403962

### **Oral Presentations**

1. **Pfaff, M. R.**, Wague, A., Davies, M., Killaars, A. R., Ning, D., Garcia, S., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Viscoelastic HyA hydrogel promotes recovery of muscle quality and vascularization in a murine model of delayed rotator cuff repair. Oral presentation presented at: Biological Scaffolds for Regenerative Medicine (BSRM); May 2025; Napa, CA.
2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Oral presentation presented at: Biomedical Engineering Society (BMES) Annual Meeting; October 11–14, 2023; Seattle, WA.
3. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Oral presentation presented at: Society For Biomaterials Annual Meeting & Exposition, "Biomaterials for Regenerative Engineering"; April 21, 2023; San Diego, CA.
4. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Oral presentation presented at: Materials Research Society 2020 Spring/Fall Meeting and Exhibit; November 28, 2020; Virtual.
5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels reinforced with field-responsive hydroxyapatite exhibiting high strength and stiffness while fully hydrated. Oral presentation presented at: Northeastern Regional Student Conference of the Society for Experimental Mechanics; June 22, 2020; Dartmouth, MA.

### **Poster Presentations**

1. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Tunable semi-synthetic hyaluronic acid hydrogels promote muscle regeneration after rotator cuff injury. Poster session presented at: ORS 50th International Musculoskeletal Biology Workshop; July 22–27, 2023; Midway, UT.
2. **Pfaff, M. R.**, Killaars, A. R., Ning, D., Davies, M., Nguyen, A., Nuthalapati, P., Liu, M., Liu, X., Feeley, B. T., Healy, K. E. Semi-synthetic hyaluronic acid hydrogels promote muscle regeneration by inducing BAT differentiation of FAPs. Poster session presented at: Society For Biomaterials (SFB) Annual Meeting & Exposition; April 2023; San Diego, CA.
3. **Pfaff, M. R.**, Erb, R. M. Understanding rod rotation in fiber reinforced hydrogels: a potential toughening mechanism. Poster session presented at: Research, Innovation, and Scholarship Expo; April 15, 2021; Virtual.

4. **Pfaff, M. R.**, Alander, C. B., Hansen, M. F. Synthetic lethal approaches to treating osteosarcoma. Poster session presented at: University of Connecticut Health Center Summer Research Fellowship Program; July 27, 2017; Farmington, CT.
5. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: Research, Innovation, and Scholarship Expo; April 13, 2017; Boston, MA.
6. **Pfaff, M. R.**, Faust, J. L., Erb, R. M. Dual-network hydrogels with functionalized CaP rods for bone graft applications. Poster session presented at: National Conference on Undergraduate Research; April 6–8, 2017; Memphis, TN.

## **FELLOWSHIPS & AWARDS**

|                                                                                               |      |
|-----------------------------------------------------------------------------------------------|------|
| National Science Foundation Graduate Research Fellowship (\$138,000)                          | 2021 |
| Research, Innovation, and Scholarship Expo (RISE) Excellence in Research Award (\$2,000)      | 2021 |
| PEAK Summit Award, undergraduate research grant (\$3,000)                                     | 2020 |
| PEAK “Shout It Out!” Award, conference travel grant                                           | 2020 |
| RISE Innovation Alley Selection (poster selected for highest impact and commercial potential) | 2017 |

## **LEADERSHIP & SERVICE**

### **Food Not Bombs**

**September 2024 – Present**

- Lead a weekly serving shift for Food Not Bombs East Bay, picking up meals prepared by a team of volunteer cooks and serving 25–40 free vegetarian meals per shift to unhoused and low-income community members
- Transformed an unstaffed curbside food drop-off into an in-person served meal, bringing a table and serving each meal directly to create a more personal experience and build ongoing relationships with community members

### **Sigma Xi, The Scientific Research Honor Society (Associate Member)**

**March 2020 – Present**

Vice President, UC Berkeley Chapter

2021 – Present

- Co-organize the chapter’s semi-annual Grants in Aid of Research (GIAR) program, which awards research grants of up to \$750 to undergraduate, master’s, and PhD students across STEM
- Founded and chair the Committee for Student Advancement (CSA), growing it to as many as 40 students, including preliminary members not yet eligible for associate membership
- Host networking events and student and guest talks, including a talk by the Editor-in-Chief of *American Scientist*
- Independently planned and executed the Sigma Xi–ATP Bio Research Symposium & Poster Session, open to students at all levels, handling everything from securing a funding partner to recruiting presenters and day-of logistics; drew ~25 presenters, a keynote speaker, and ~150 attendees who voted for a best-poster prize
- Lead chapter recruitment through outreach to campus organizations and departments, and manage chapter operations, including meeting scheduling, event logistics, and records

Diversity & Inclusion Chair, Northeastern University Chapter

2020 – 2021

- Expanded outreach to first-generation, culture-specific, and other underrepresented student organizations, and hosted a campus-wide diversity, equity, and inclusion panel

### **BioEngineering High School Competition (BioEHSC)**

**April 2022 – April 2026**

- Academic Judge: evaluated 10–15 high school team pitches each spring in a full-day bioscience pitch competition, assessing depth of knowledge through Q&A and providing feedback

### **Northeastern University Club Sports**

**September 2016 – June 2021**

- Triathlon Team: President/Captain (2019–2020), Chief Financial Officer (2018–2019), and Chief Procurement Officer (2017–2018)
- Club Sports Council (2019–2021): served as liaison between club sports teams and Club Sports administrators and directors

**Adventure Challenge Volunteer****September 2018 – December 2019**

- Planned and led weekly swimming and pool games with a Boston Children's Hospital patient with a disability to foster movement, practice social skills, and build connection

**Northeastern University Resident Assistant****August 2017 – July 2019**

- Served as a live-in mentor and resident advisor for ~45 first-year women a year from around the world pursuing careers in STEM, helping them set and achieve goals, connect with scientists and professors, and navigate issues including academic performance, inclusion, and substance abuse
- Designed and implemented educational, recreational, and networking programs, including Q&A panels with graduate students and professors, leadership and team-building workshops, and faculty mentorship events

**Advancement Via Individual Determination (AVID) Tutor****September 2018 – December 2018**

- Led a weekly group of 6–8 students at Fenway High School in the AVID college-readiness program, teaching collaborative problem-solving and critical thinking for math and science

**CLINICAL EXPERIENCE****Tufts Medical Center, Oncology and Hematology****Boston, MA**

Clinical Care Technician

January 2019 – December 2019

- Provided direct care for 6–10 oncology and hematology patients per shift, including support with mobility, hygiene, and meals
- Monitored vital signs and blood glucose, escalated abnormal findings to nursing staff, and supported clinical teams during examinations, procedures, and emergencies

**Fallon Ambulance Service****Quincy, MA**

Emergency Medical Technician Basic

September 2018 – April 2019

- Assessed, treated, and transported ill and injured patients, providing immediate medical and psychological care en route to hospitals and specialty facilities
- Documented vitals, symptoms, medical history, allergies, and medications per company protocols while maintaining a calm, safe environment for patients and families

**Dental Office of Dr. Andrew J. Ponichtera DMD, MS****Simsbury, CT**

Dental Assistant

June 2016 – August 2016

- Supported chairside procedures including fillings, root canals, and placement of crowns, implants, bridges, and dentures
- Managed front-desk operations and supply inventory, including patient scheduling, records, and ordering

**SKILLS****Biomaterials Synthesis & Mechanical Testing**

- Hydrogel synthesis and chemical modification, drug loading and release, rheology, injectability testing, mechanical testing (dynamic mechanical analysis, tensile, burst, and lap shear), freeze casting, lyophilization

**Materials Characterization & Microfabrication**

- NMR spectroscopy, X-ray photoelectron spectroscopy (XPS), scanning electron microscopy (SEM), microfabrication and photolithography

**Cell & Molecular Biology**

- Primary and 3D cell culture, immunocytochemistry, confocal and fluorescence microscopy, histology, flow cytometry, qPCR/ddPCR, Western blot, molecular cloning, CRISPR-Cas9 gene editing (transfection, nucleofection, viral transduction), automated liquid handling

**Data Analysis & Computation**

- R and Python (including single-cell RNA-seq analysis), GraphPad Prism, MATLAB, COMSOL Multiphysics (finite element analysis)

**Regulatory & Translational**

- FDA regulatory pathways and product classification, environmental health and safety (EHS) compliance, laboratory buildout and procurement, grant writing